

SERTAR, China

WMO: 561520

Lat: 32.2749N

Lon: 100.3337E

Elev: 3896

StdP: 62.48

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.0	-20.0	-29.1	0.4	-14.0	-27.3	0.5	-13.0	11.8	-0.2	10.4	0.2	0.8	290	0.724

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	19.7	10.7	18.5	10.1	17.2	9.6	11.9	17.7	11.2	16.7	10.6	15.7	2.5	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.8	12.3	14.4	9.2	11.8	13.6	8.6	11.4	13.0	47.8	17.8	45.4	17.0	43.4	15.9	16.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.3	7.7	6.4	DB	-25.8	22.3	2.0	0.8	-27.2	22.9	-28.4	23.4	-29.5	23.8	-30.9	24.5	(4)
(5)			WB	-26.2	13.3	1.8	1.0	-27.5	14.0	-28.6	14.6	-29.6	15.2	-31.0	15.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	2.0	-8.8	-6.1	-2.4	2.3	6.2	9.7	11.2	10.7	8.4	3.1	-3.5	-7.8	(6)
(7)		DBStd	7.63	3.04	3.03	2.94	2.48	2.62	2.48	2.06	2.05	2.60	3.15	3.20	2.93	(7)
(8)		HDD10.0	3063	583	451	385	232	120	36	12	17	60	213	404	551	(8)
(9)		HDD18.3	5974	841	684	643	482	374	259	221	237	298	471	654	809	(9)
(10)		CDD10.0	131	0	0	0	0	4	26	51	38	12	1	0	0	(10)
(11)		CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)
(12)		CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)
(13)		CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	2.1	2.3	2.5	2.5	2.4	2.3	2.1	1.9	1.7	1.7	1.8	1.9	2.0	(14)
(15)	Precipitation	PrecAvg	649	4	7	13	23	60	133	144	103	117	34	6	4	(15)
(16)		PrecMax	802	16	18	37	50	103	219	254	175	180	82	20	12	(16)
(17)		PrecMin	515	0	1	0	5	13	59	57	32	41	6	0	0	(17)
(18)		PrecStd	67	4	5	9	9	21	36	49	36	36	18	5	4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.2	9.4	11.6	15.6	19.1	20.6	21.2	21.2	19.7	17.2	10.7	7.9	(19)
(20)			MCWB	-1.7	-1.1	1.0	3.5	7.6	10.7	11.9	11.2	10.0	7.0	0.2	-1.6	(20)
(21)		2%	DB	4.6	6.7	9.0	12.9	16.8	18.2	19.6	19.5	17.8	13.9	8.3	5.3	(21)
(22)			MCWB	-3.6	-2.4	-0.3	2.6	6.5	10.0	11.1	10.8	9.5	5.8	-0.8	-3.0	(22)
(23)		5%	DB	2.3	4.6	7.0	11.1	14.8	16.7	18.2	18.0	15.9	11.5	6.4	3.0	(23)
(24)			MCWB	-4.8	-3.2	-1.2	1.7	5.6	9.6	10.7	10.3	9.1	4.3	-1.5	-4.1	(24)
(25)		10%	DB	0.1	2.6	5.2	9.1	12.9	15.2	16.7	16.4	14.2	9.3	4.4	0.9	(25)
(26)			MCWB	-5.9	-4.2	-2.0	1.1	5.0	8.8	10.2	9.8	8.2	3.2	-2.4	-5.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.4	-0.9	1.7	4.9	9.2	12.2	13.2	12.6	11.4	8.2	0.9	-1.2	(27)
(28)			MCDWB	7.3	8.1	8.8	11.8	16.5	18.0	19.2	18.7	17.5	15.2	7.8	7.1	(28)
(29)		2%	WB	-3.3	-2.0	0.3	3.7	7.9	11.1	12.2	11.6	10.3	6.6	-0.2	-2.8	(29)
(30)			MCDWB	4.1	5.8	7.2	10.5	14.8	16.8	17.7	17.6	16.0	12.5	6.8	4.4	(30)
(31)		5%	WB	-4.7	-3.0	-0.7	2.7	6.8	10.3	11.5	11.0	9.5	5.2	-1.1	-3.9	(31)
(32)			MCDWB	1.7	4.0	5.5	8.9	12.9	15.6	16.7	16.6	14.8	9.8	5.2	2.5	(32)
(33)		10%	WB	-5.8	-4.0	-1.6	1.8	5.8	9.4	10.8	10.3	8.7	4.0	-2.0	-5.1	(33)
(34)			MCDWB	-0.3	1.7	4.1	7.6	11.1	14.1	15.6	15.3	13.5	7.9	3.6	0.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	18.5	17.0	14.5	12.3	11.9	10.2	10.6	11.3	11.5	12.0	15.6	18.2	(35)
(36)			MCDBR	22.0	21.1	18.0	16.8	15.9	12.6	13.9	14.9	14.3	15.2	18.4	21.3	(36)
(37)			MCWBR	14.6	13.2	9.8	7.7	6.8	5.6	6.4	7.0	6.9	7.4	10.2	13.8	(37)
(38)		5% WB	MCDBR	20.6	19.2	15.4	13.3	12.7	10.8	10.9	12.0	12.4	12.8	16.4	20.3	(38)
(39)			MCWBR	13.9	12.2	8.5	6.5	5.9	5.2	5.3	5.8	6.2	6.5	9.4	13.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.212	0.240	0.290	0.309	0.300	0.288	0.289	0.288	0.264	0.239	0.213	0.198	(40)	
(41)		taud	2.588	2.510	2.348	2.341	2.451	2.596	2.613	2.597	2.670	2.668	2.652	2.657	(41)	
(42)		Ebn at Noon	1048	1035	992	979	984	990	988	989	1011	1026	1040	1054	(42)	
(43)		Edh at Noon	77	93	119	125	114	98	96	96	85	78	71	68	(43)	
(44)	All-Sky Solar Radiation	RadAvg	3.83	4.67	5.74	6.51	6.06	5.42	5.61	5.33	4.72	4.20	4.08	3.69	(44)	
(45)		RadStd	0.28	0.23	0.29	0.53	0.46	0.46	0.38	0.57	0.36	0.23	0.21	0.20	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	+0.52	+1.07	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47)	Regional (5 neighbors)	+0.40	+0.56	-0.81	+0.63	N/A	N/A	-104	-168	+61	N/A		

Nomenclature: See separate page